

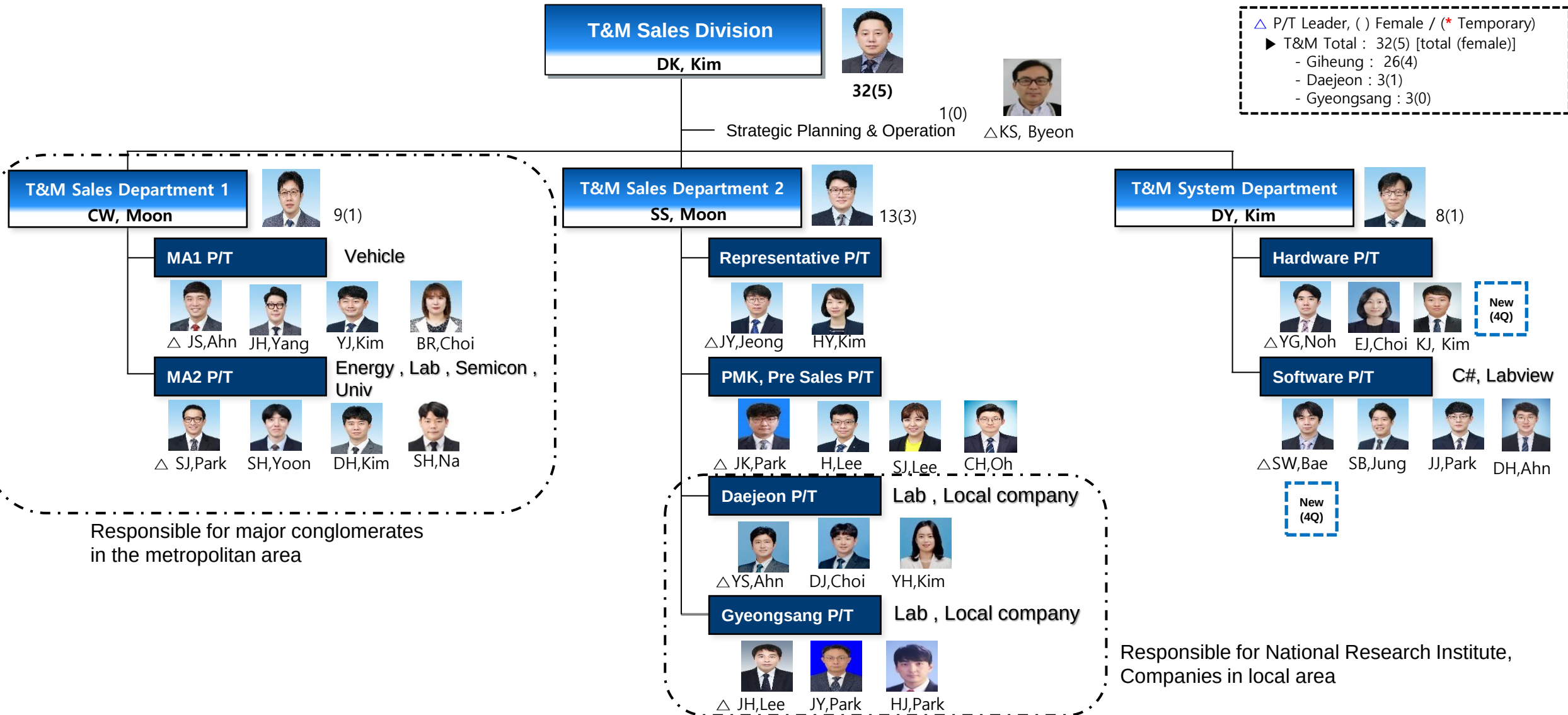


Top Gun Project DL950 Success Stories

2025-02-18

Confidential

YKO T&M Dept. Organizations



Self introduction



Profile

Name : BAE Seong woo(裴成友)

E-mail : seongwoo.bae@yokogawa.com

Birthday : 1980.05.03

Place of birth : Busan, Korea

Work(Assigned Task)

Develop programs to implement customer systems or services

Development Language : C++, C#, JAVA

Career

Power Noise Simulator

[Hyundai mobis](#), [Samsung SDI](#)

Battery TEST system

[Samsung SDI](#) – DL9040

Camera durability test system

[Hyundai mobis](#), [LG Innotek](#)

Thermoelectric module and Peltier element efficiency and durability tester

[Hyundai](#)

Grid PV Simulator

[Hanwha](#) – WT1800, WT3000, WT5000

DL950 Success Stories

Client



YURA TECH

auto parts manufacturer

(Major customers are **Hyundai/Kia Motors and Hyundai Mobis.**)

DL950 Sales volume – 6EA

Spark plug System

DL950-F-HK - 5EA

720211 20MHz -15EA

701933 30A 50MHz – 5EA

Order amount 250,000,000 won

Pre-charge Relay System

DL950-F-HK/M1 - 1EA

720268 1MS/s - 5EA

720256 4CH 10MS/s 12Bit Isolation Module - 1EA

VZ20X -1EA

Order amount 123,000,000 won

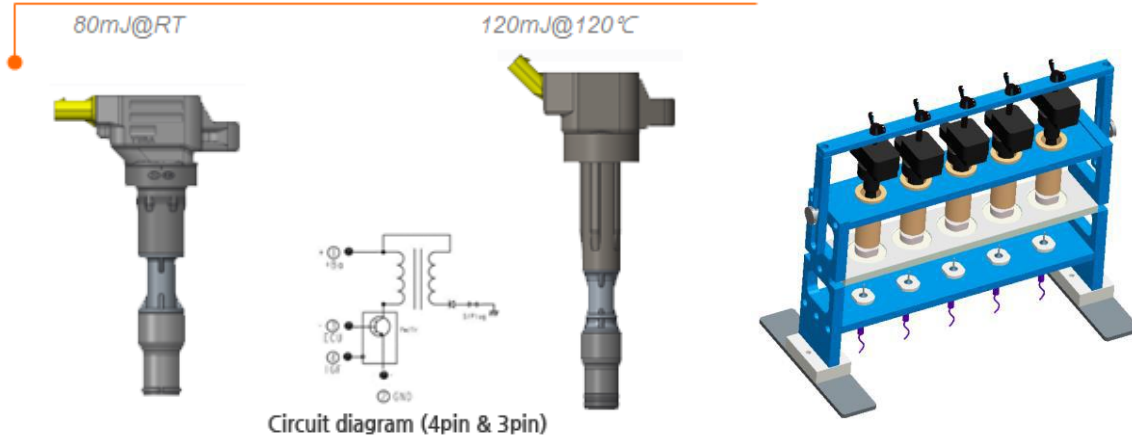
DL950 Success Stories

Spark plug durability System

Customer Requirements

- Load resistance: 10Mohm
- Maximum voltage: 50kV
- Voltage and current level measurement while operating Spark plug 30CH.
- Test data acquisition for 6 months (They must provide the test data to Hyundai Motor)

Plug Top Type

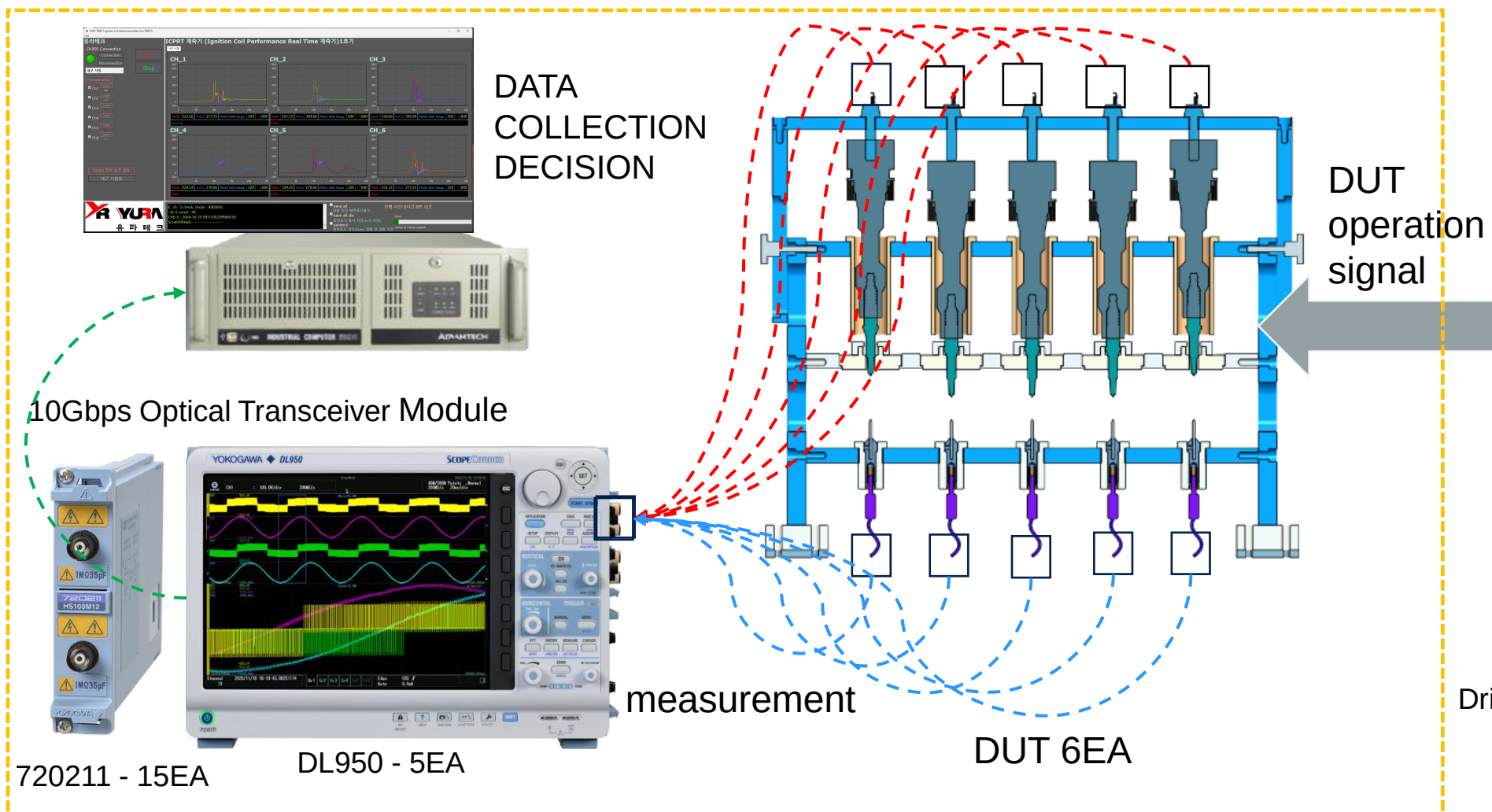


The core driving device of an engine that generates high-voltage current to cause spark discharge inside the engine.

DUT - Spark plug

DL950 Success Stories

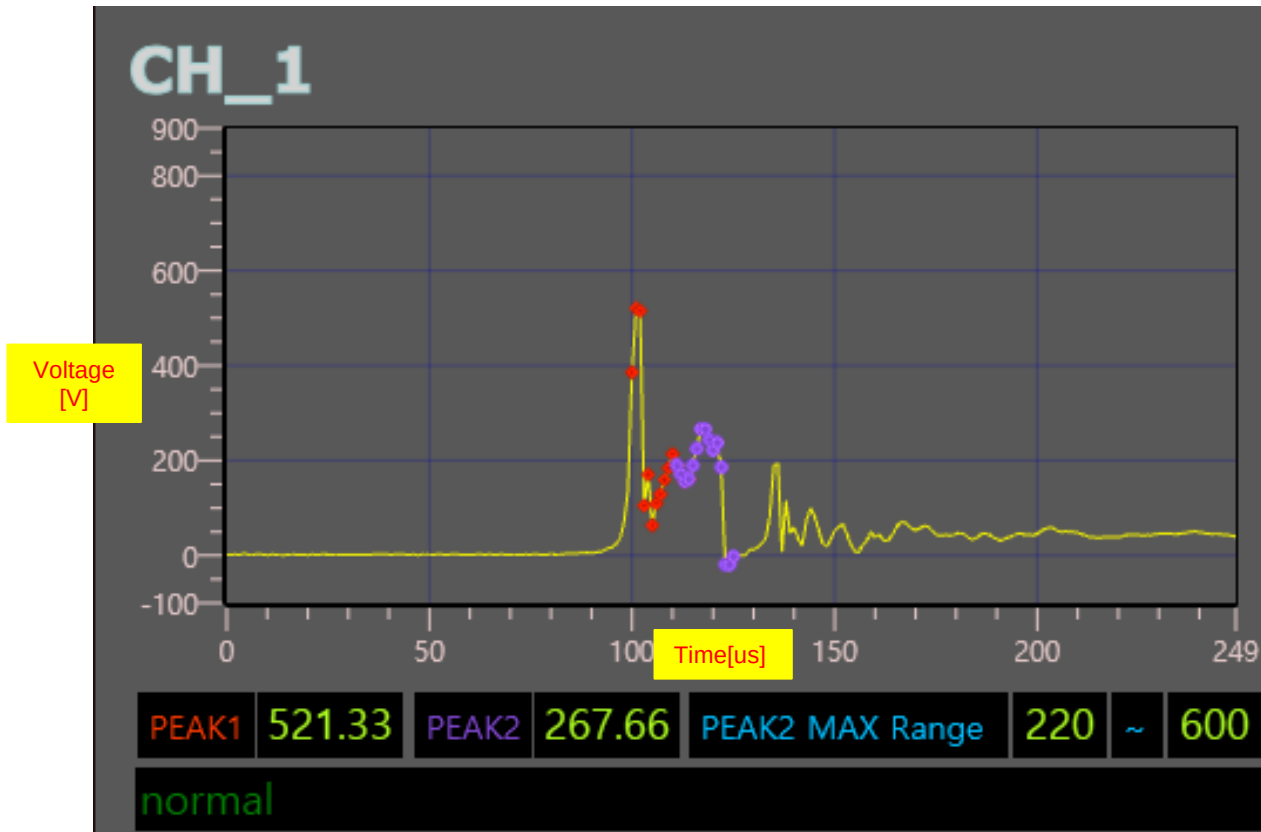
Spark plug durability System - Configuration



Drive units delivered in the past

DL950 Success Stories

Spark plug durability System – S/W Program



Waveform display(2M Sampling, Point-Point 2us)

1st peak measurement(red dot)

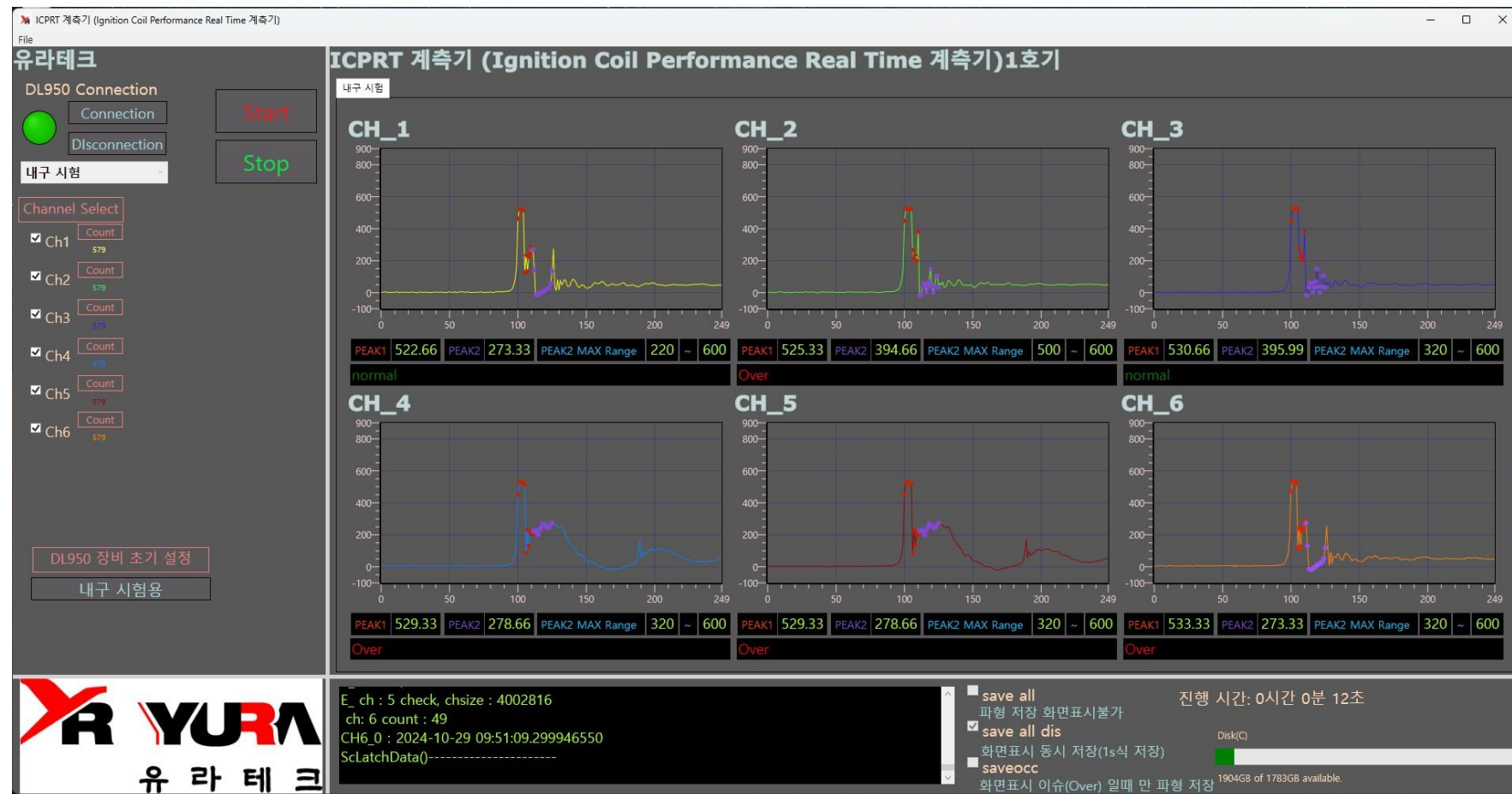
2nd peak measurement(purple dot)

2nd peak - Reading by range

- Normal range "normal" display
- "Over" display outside normal range
- 2nd peak Range: User configurable

DL950 Success Stories

Spark plug durability System - Drive photos



DL950 Success Stories

Spark plug durability System - On-site photos



DL950 Success Stories

Pre-charge Relay System

Customer Requirements

Data monitoring & save for electronic characteristics

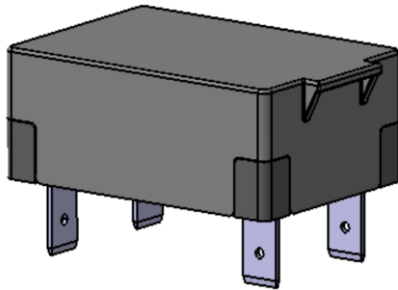
Cycle operation On/Off for DUT

Alarm function for overcurrent & high temperature

DUT control according to temperature and humidity chamber pattern.

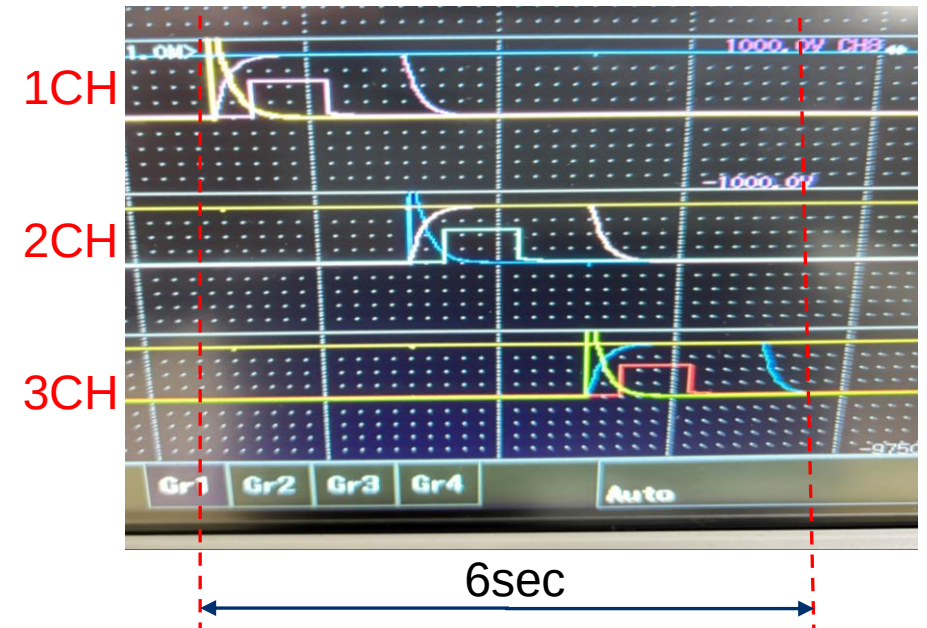
The condition is that the 3CH operation is completed within 6sec.

DUT – Pre-Charge Relay



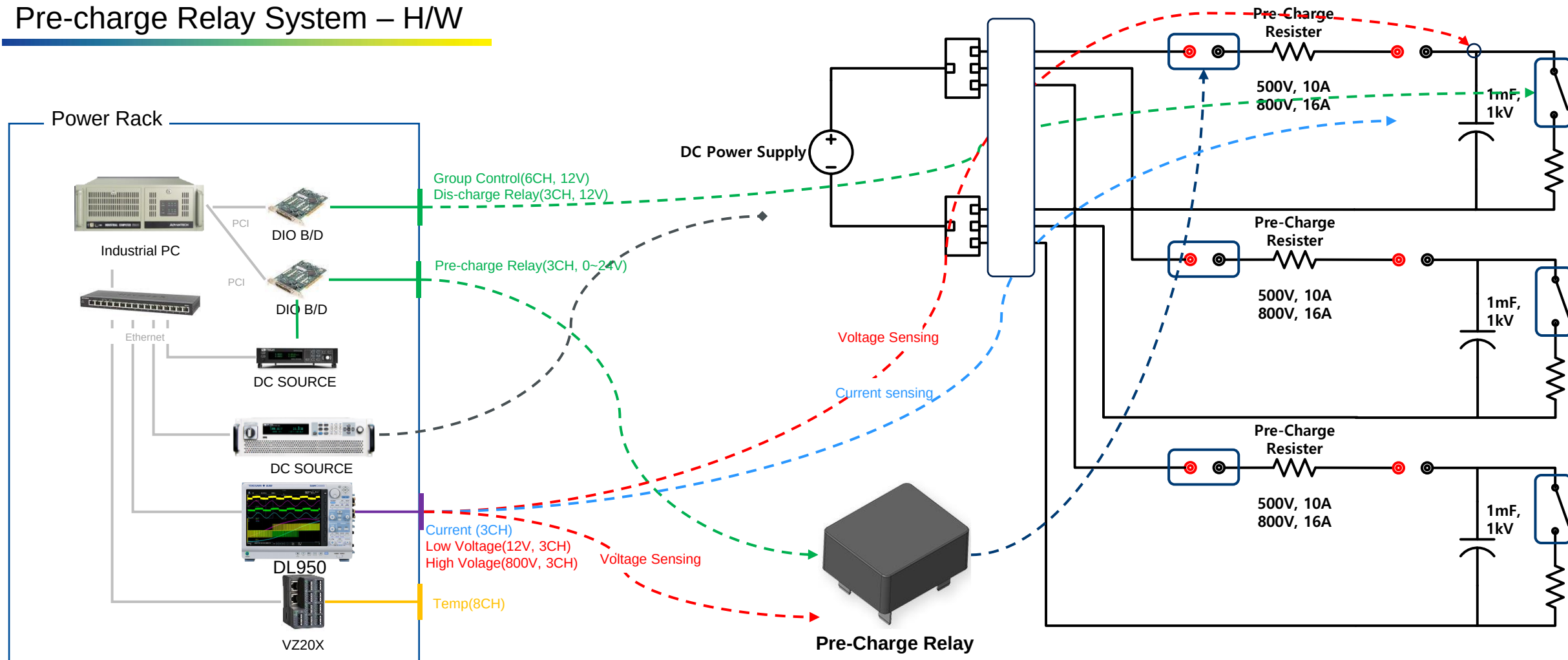
Components that connect and blocks electric vehicle (EV) batteries and inverters/chargers

DL950 screen



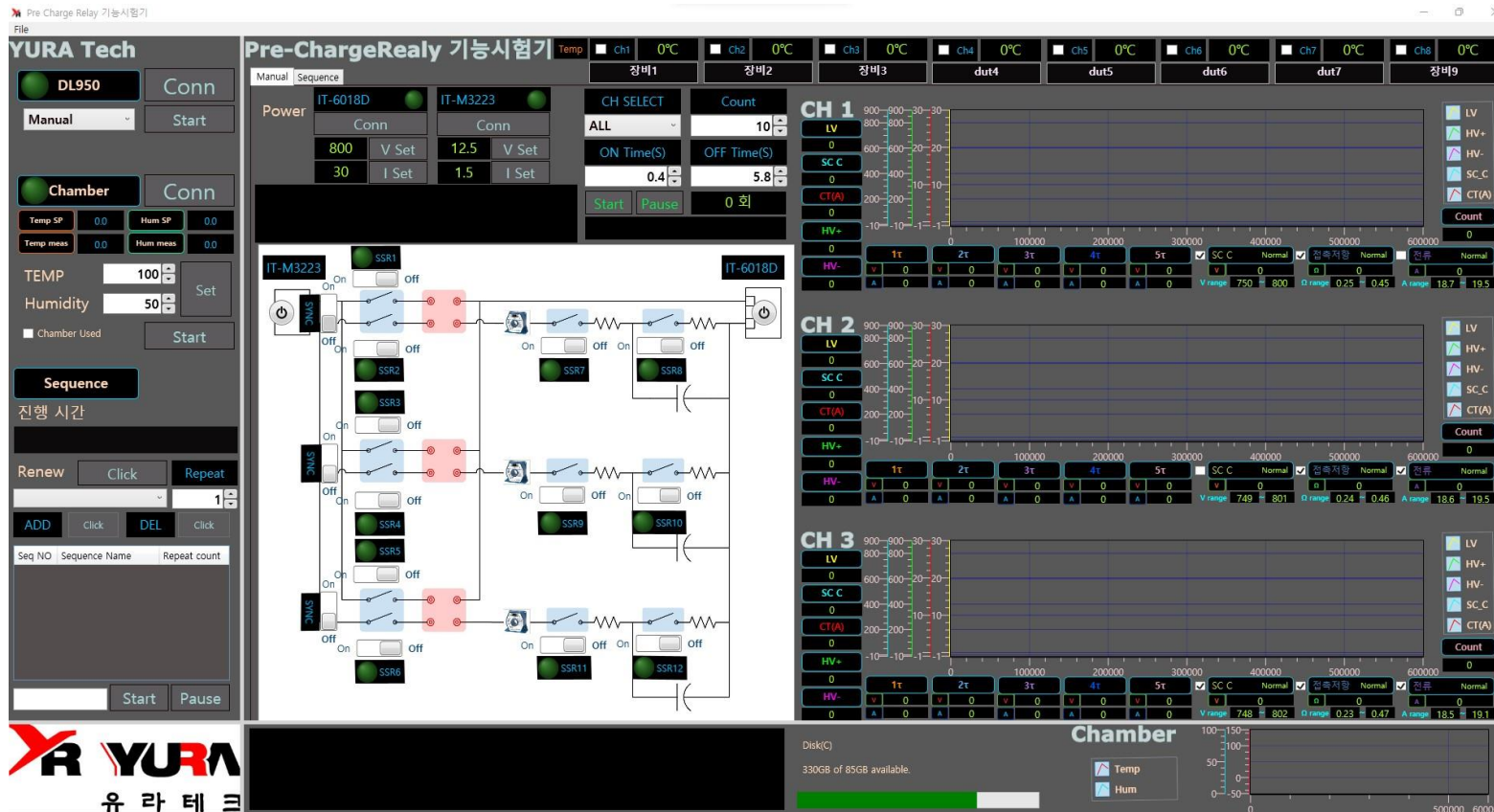
DL950 Success Stories

Pre-charge Relay System – H/W



DL950 Success Stories

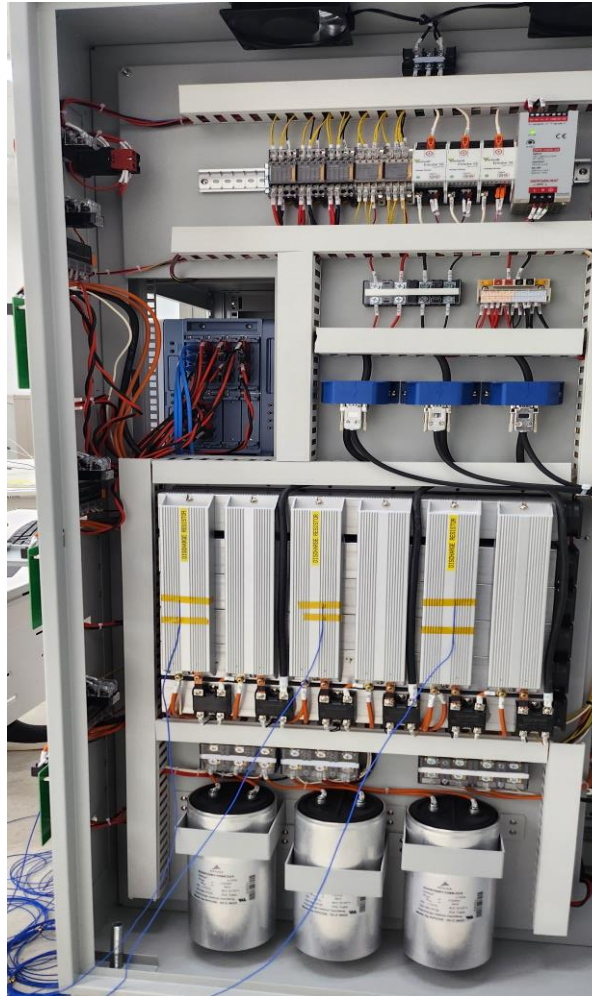
Pre-charge Relay System – S/W Program



- ON OFF TIME Control Operation
- Chamber Control
- Tau(Time constant) Calculation
- Contact resistance Calculation

DL950 Success Stories

Pre-charge Relay System - On-site photos



Self introduction



Profile

Name : Jung Seungbae(丁陞陪)
E-mail : seungbae.jung@yokogawa.com
Birthday : 1982.01.23
Place of birth : Yangsan-si,
Gyeongsangnam-do, Korea

Work(Assigned Task)

Programs Developer for PC Based
Application

Development Tools : LabVIEW, C#

Career

Pre-Charge Relay Tester

[Woojin](#)
(DL950 3EA)

Li-ion Battery Charge/Dis-Charge Tester

[KyungshinHoldings](#)
(WT300 Series 15EA, GM10 3EA)

EOL Tester for BDU Function

[YURA Corporation](#)

EOL Tester for ICCB Function tester

[YURA Corporation](#)
(DLM3000 Series 8EA)

■ Customer Information



Company title: Woojin Industry Company Ltd.

Establishment: May, 1966

Customers: **Hyundai, Kia, GM**, etc.

Products: Spark Plug, Oxygen Sensor, Glow Plug, etc.

Annual Sales: 218.66599 billion won

→ Oder amount(for Top gun): 615 million won

DL950 Success Stories

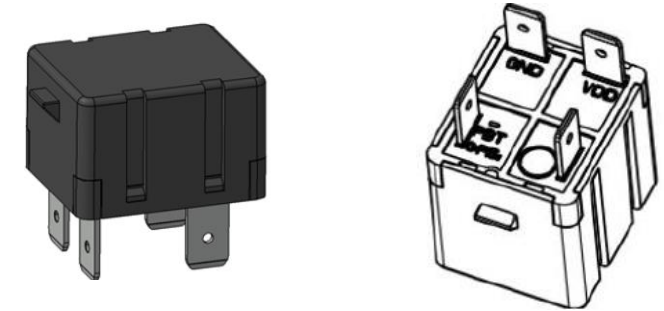
DUT(Device Under Test) Specifications

EPR, Electronic Pre-charge Relay

Item	Spec.	Remark
Coil Voltage	8~16Vdc	Max. 650 Vdc(Output)
Current Rating(Output)	11.25A	450Vdc, 40Ω, 1000uF(Output)
Pick Up Voltage	8Vdc Below	
Drop Out Voltage	3.1 +/- 0.5Vdc	12Vdc, 1kΩ @ 20°C(Output)
Operate Time	0.1 ~ 1.0ms	
Release Time	0.8 ~ 3.8ms	
Pickup current	5 ~40ms	
Contact Resistor(Output)	0.5 ~ 1.2Ω	12Vdc, 1A @ 20°C(Output)

Test Specification

[Durability test](#), [Environment reliability test](#) and [Mechanical reliability test](#) for EPR
HYUNDAI MOTOR Engineering Standard



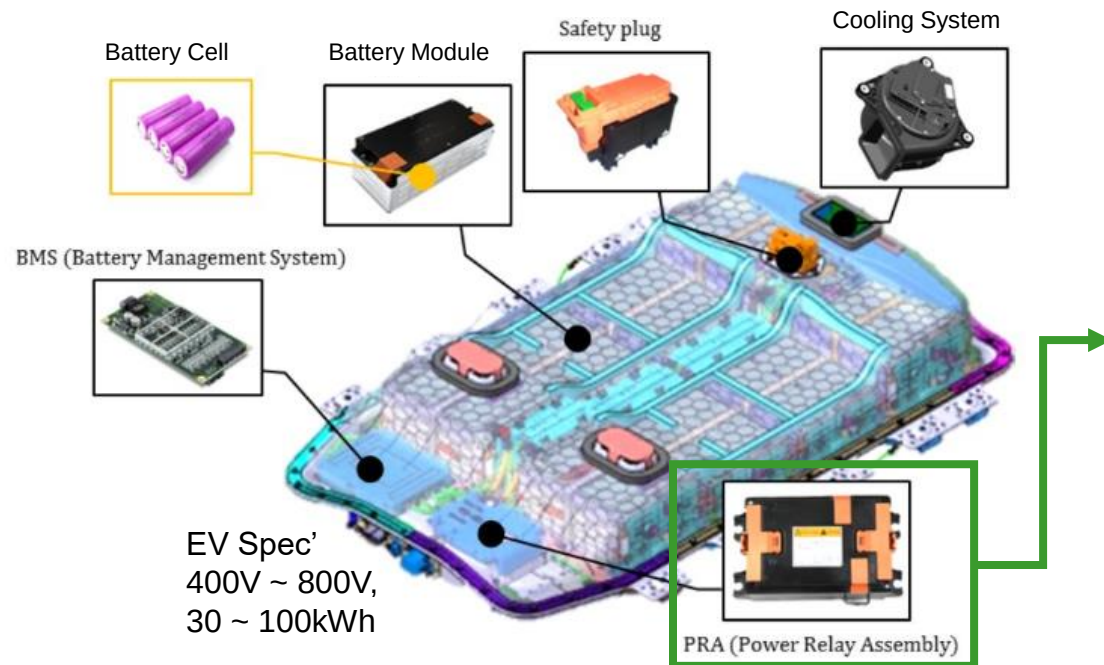
EPR

As a PRA SUB part, it operates before the main relay operates and reduces the voltage difference within the circuit to prevent main relay overload.



DL950 Success Stories

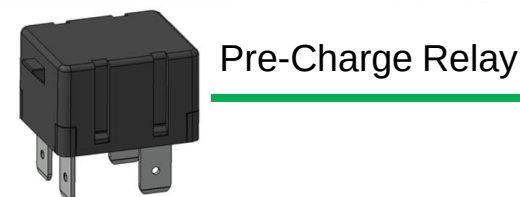
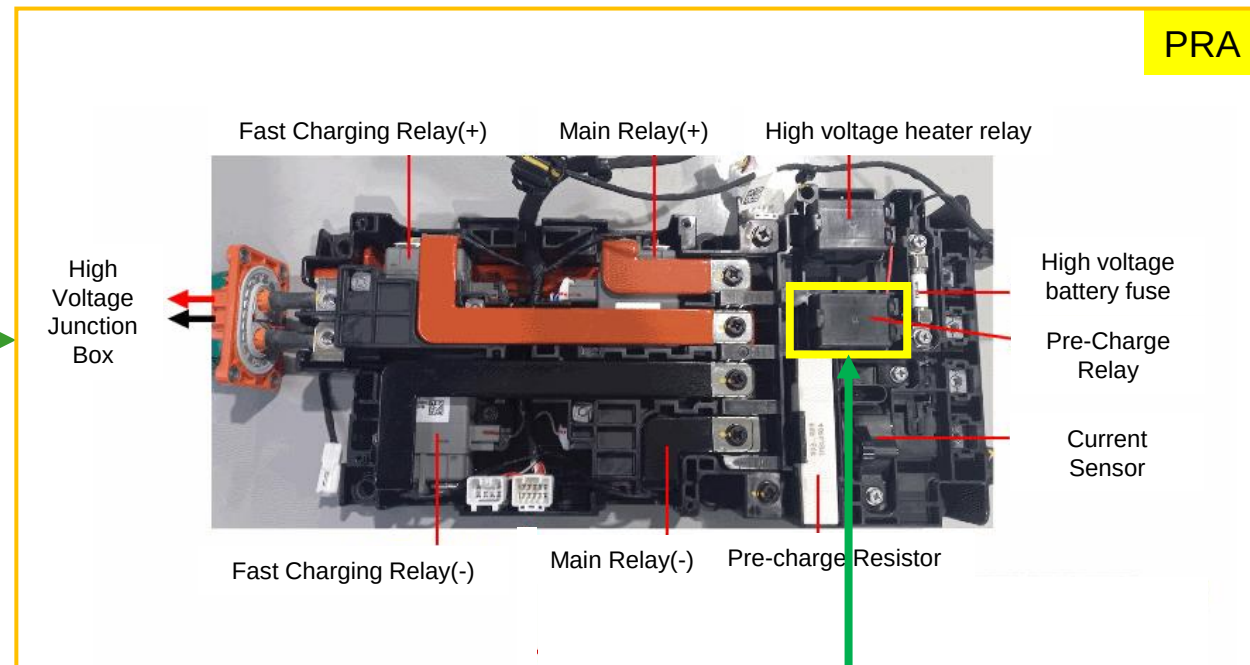
What is the EPR?



The **Power Relay Assembly (PRA)** is a component that supplies and cuts off power for discharging or charging the high-voltage battery.

It is installed in the high-voltage battery system assembly (BSA) and electrically connects the battery and the inverter.

Applicable components include main relay, pre-charge relay, resistor, current sensor, temperature sensor, bus bar, etc.



As a PRA SUB part, it operates before the main relay operates. reduces the voltage difference within the circuit to prevent main relay overload.

100

720268 High Voltage Module 1MS/s



DSP1950



701265 High-Precision Temperature Input Module

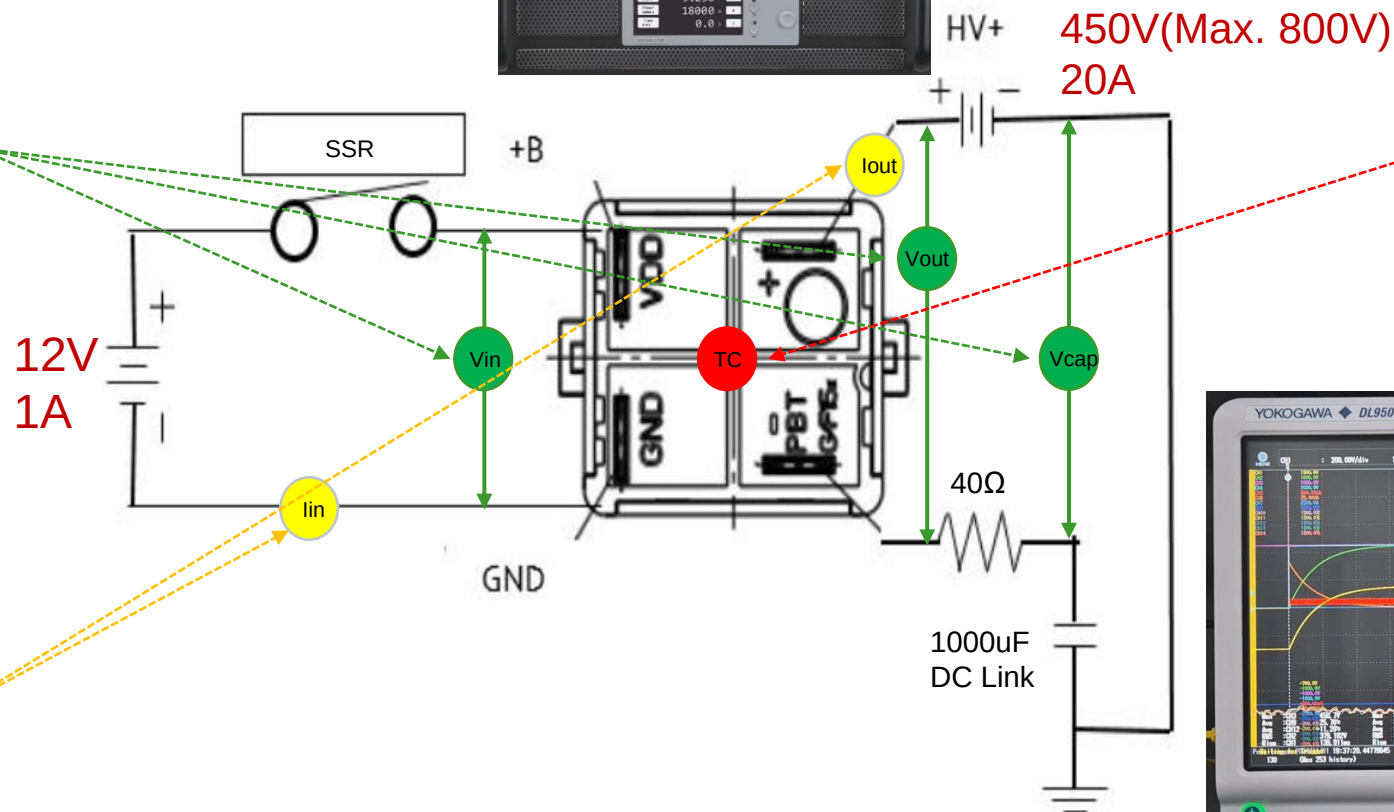


BP4610

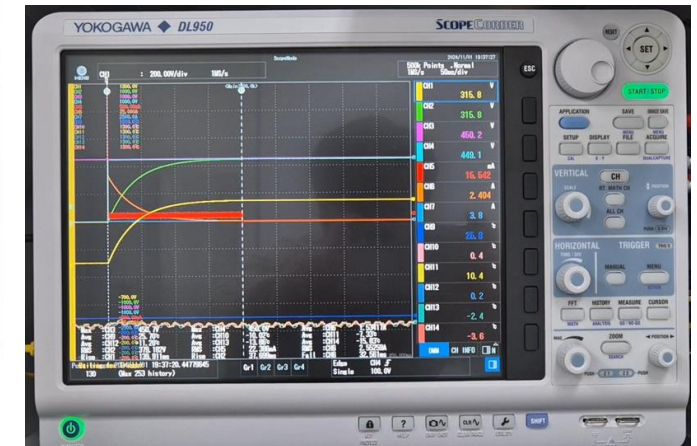


12V
1A

701251 Analog Voltage Input Module

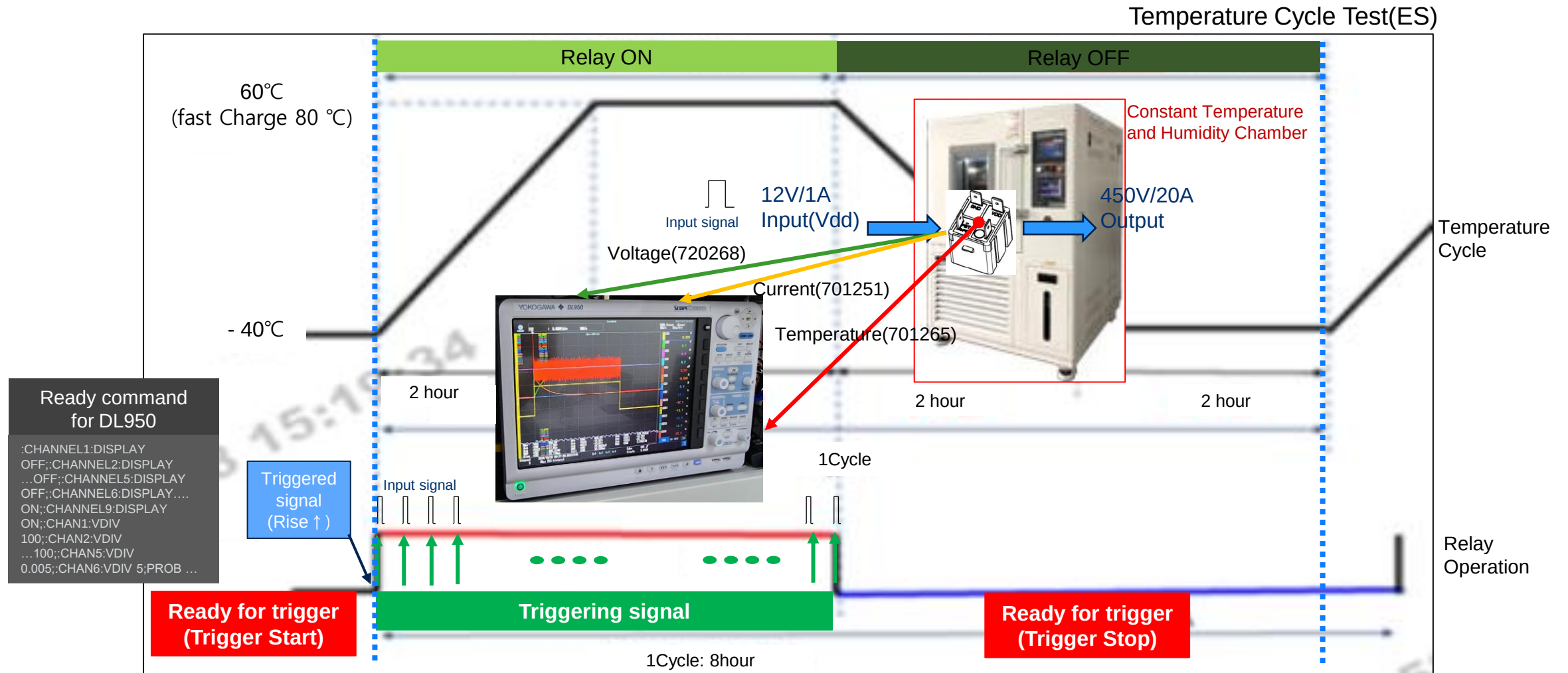


DL950



DL950 Success Stories

DL950 Solution for ES Test of High Voltage Relay



Ready command
for DL950

```
:CHANNEL1:DISPLAY  
OFF::CHANNEL2:DISPLAY  
...OFF::CHANNEL5:DISPLAY  
OFF::CHANNEL6:DISPLAY...  
ON::CHANNEL9:DISPLAY  
ON::CHAN1:VDIV  
100::CHAN2:VDIV  
...100::CHAN5:VDIV  
0.005::CHAN6:VDIV 5;PROB ...
```


Mechanical & Environment Test, 2EA

Test System



DUT 3EA
Switching

DL950 Modules used

720268 High Voltage Module
1MS/s



X 2

701265 High-Precision Temperature
Input Module



X 2

701251 Analog Voltage Input Module



X 1

702916 Current Probe 120 MHz/ 0.5
ARMS, 5 ARMS, 30 ARMS



X 1

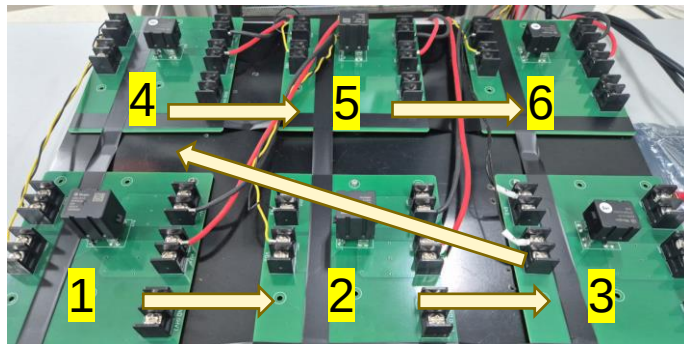
Durability Test, 1EA

Test System



DUT
6EA

Switching



DL950 Modules used

720268 High Voltage Module
1MS/s



X 2

701265 High-Precision Temperature
Input Module



X 3

701251 Analog Voltage Input Module



X 1

702916 Current Probe 120 MHz/ 0.5
ARMS, 5 ARMS, 30 ARMS



X 1

For 80A Test

701930 Current Probe,
150Arms,DC to 10MHz



X 1

DL950 Success Stories

System solutions for installed customer sites

[DL950 Movie](#)



Expectations for DL950 Application

T&M System

It is expected to be a device that can replace NI DAQ products based on noise processing and isolation.

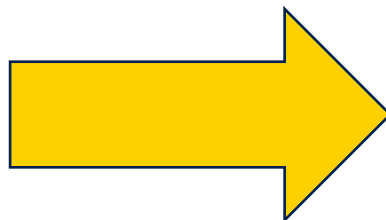


NI compactDAQ



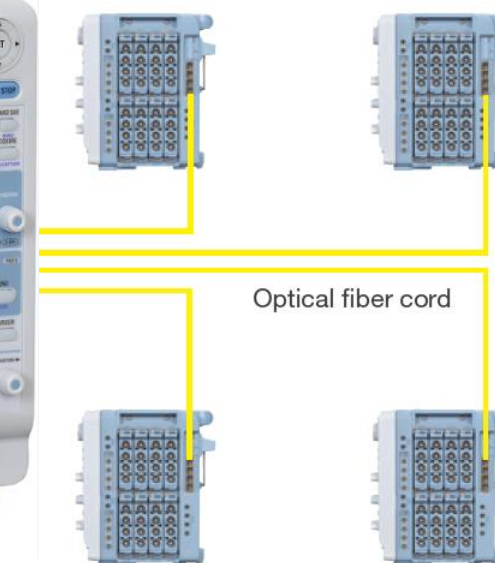
NI PXI

Replacement for
Test Solution



Applications

- Battery cell evaluation
- Multi-point vibration analysis
- Multi-point strain test



Optical fiber cord

*Please use the Optical Transceiver Module 720941 and the Optical Fiber Cord 720942.

Sales opportunity of DL950

■ Samsung MX Division (Mobile)

- 1unit
- Circuit board duration & current test

■ Stats Chip pack Korea

- 1unit
- Semiconductor duration & looping test

■ Posco

- 2 units
- Electricity quality monitoring for electrical substation in steel plant

■ Keyang Electric Machinery

- 1 unit
- Vehicle sunroof motor test

Request for YMI Product (Sales)

■ Physical Layer Analysis

- CAN/CANFD or other serial signal physical layer decoding function like DLM series

■ Integrated Printer

- Integrated printer support instantaneous data output to tangible paper
- Heavy industry customers (Shipbuilding) started to change DL950 to other high speed DAQ with printer
- The market size that can sell DL950 in the maintenance market is bigger. However, the market is being lost due to the absence of a printer. A printer function that can be set up simply and printed is needed.

■ Weight

- Too heavy to carry to the test site

Request for YMI Product (Sales)

■ USB C Interface

- USB C interface should be applied(EX : replace VGA to USB C, USB C cable to External SSD)

■ Measure values order

- Measure values order that are displayed in the screen cannot be changed
- The order should be edited by user for the better analysis

■ Integrated DC power

- Integrated DC power helps not to carry external power bank or source.

■ Display size expansion

- The screen is relatively smaller than the DLM series. Display size expansion is required.

Request for YMI Product (Engineer)

■ Remote mode key Lock

- While in remote mode entire key locked, make us difficult to analysis the trend
- At least time axis should be changeable so that trend analysis will be possible

■ Data acquisition while waiting for Trigger

- In some cases, 720221, 701261 are used in same time with voltage input modules
- While waiting trigger, there is no way to gather temperature data

■ LabVIEW Driver

- Some commands are missing in LabVIEW driver
- The driver should support most of the functions in the communication commands

Co-innovating tomorrow™